

ShopEz

Full Stack MERN E-Commerce Application

PROJECT TECHNICAL DOCUMENTATION

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Git link: <https://github.com/m-alladi2006/VIP-C2-SHOPEZ-E-commerce-Application.git>
Team ID: _____
Date: 18 June 2026

1. Introduction

Project Title: ShopEz

Developer: Alladi Manjunadh (Solo Project)

ShopEz is a high-performance, comprehensive e-commerce platform developed to offer a seamless and contemporary shopping experience for digital consumers. Leveraging the robustness of the MERN stack (MongoDB, Express.js, React.js, and Node.js), this application connects end-users to an extensive variety of products through an interface that prioritizes responsiveness and user engagement.

2. Project Overview

Purpose: The goal of ShopEz is to provide a reliable and expandable infrastructure for online retail. It is designed to facilitate effortless product exploration, efficient cart management, and a secure transaction pipeline. Additionally, the system includes a dedicated administrative portal empowering managers with real-time inventory control and complete oversight of the order fulfillment lifecycle.

Key Features:

- **User Module:** Features include secure credential registration, JWT-based authentication protocols, and customized user profiles.
- **Product Module:** Offers organized browsing across distinct categories (Men, Women, Kids) paired with integrated search and filtering tools.
- **Cart & Checkout:** Dynamic shopping basket with real-time updates, concluding with a specialized Cash on Delivery (COD) checkout flow.
- **Admin Dashboard:** A centralized interface for product catalog updates, financial analysis, and tracking order delivery statuses.

3. System Architecture

Frontend (React.js):

The client-side interface utilizes a Component-Based Architecture to promote code modularity and maintainability. Global application state is handled through the React Context API and localized hooks. The user experience is optimized as a Single Page Application (SPA), utilizing React Router DOM for efficient client-side navigation without full-page reloads.

Backend (Node.js s Express.js):

The server-side component employs an MVC-patterned logic to deliver a scalable RESTful API. Express.js facilitates the routing layer, while customized middleware logic ensures data integrity, rigorous authentication, and security. The backend communicates with the frontend via structured JSON payloads to ensure smooth data interchange.

Database (MongoDB):

A NoSQL document-based database approach is used for high-velocity data persistence and flexible schema management. Key database resources include:

- **Users:** Manages profile details, encrypted credentials, and access roles (User vs. Administrator).
- **Products:** Stores detailed item metadata such as names, descriptions, categories, price points, and remaining stock.
- **Orders:** Maintains comprehensive transaction records, including itemized lists and logistical tracking info.

4. Setup Instructions

Prerequisites:

- Node.js (Version 18.x or higher)
- Package Ecosystem: NPM or Yarn
- MongoDB Community Server (Configured for local port 27017)

Installation Steps:

1. Repository Initialization:

```
https://github.com/m-alladi2006/VIP-C2-SHOPEZ-E-commerce-Application.git
```

2. Server Configuration:

```
cd server && npm install
```

3. Client Configuration:

```
cd client && npm install
```

4. Environment Setup: Create a .env file in the server root folder with these keys:

```
PORT=5000  
MONGO_URI=mongodb://localhost:27017/shopez  
JWT_SECRET=your_secure_secret_key
```

5. Project Organization

Directory	Description
<code>/server/models</code>	Database schemas defined via Mongoose for Users, Products, and Order tracking.
<code>/server/routes</code>	Mapping of API endpoints to their specific controller functions and logic handlers.
<code>/client/src/components</code>	Modular UI blocks such as navigation bars, footers, and interactive product cards.
<code>/client/src/pages</code>	High-level view templates for the storefront, product views, and checkout screens. <code>c</code>

6. Launching the System

To start the application in your local development environment, run the following commands in separate terminals:

Backend Service:

```
cd server && npm run dev
```

Frontend Application:

```
cd client && npm start
```

7. API Specifications

Method	Endpoint	Functionality
POST	/api/users/login	Authenticates visitor credentials and provides a secure JWT.
GET	/api/products	Fetches the full product list with support for filtered searches.
POST	/api/orders	Processes and records a new COD transaction (Auth required).

8. Security: Auth s Permissions

The ShopEz ecosystem utilizes JSON Web Tokens (JWT) for maintaining secure, stateless visitor sessions. Following a verified login, the server generates a signed token stored within the client's LocalStorage or browser cookies. This token is attached to the Header of subsequent protected requests. Custom authorization middleware validates these tokens and enforces role-specific permissions, particularly for accessing sensitive administrative features.

9. UI Design Philosophy

The frontend layout follows a minimalist and clean aesthetic to ensure maximum usability. Core components include a responsive sticky navigation bar, grid-style product visualizations, and a logical checkout sequence. High-resolution graphic placeholders and thematic banners are utilized throughout to maintain a professional look that aligns with modern retail standards.

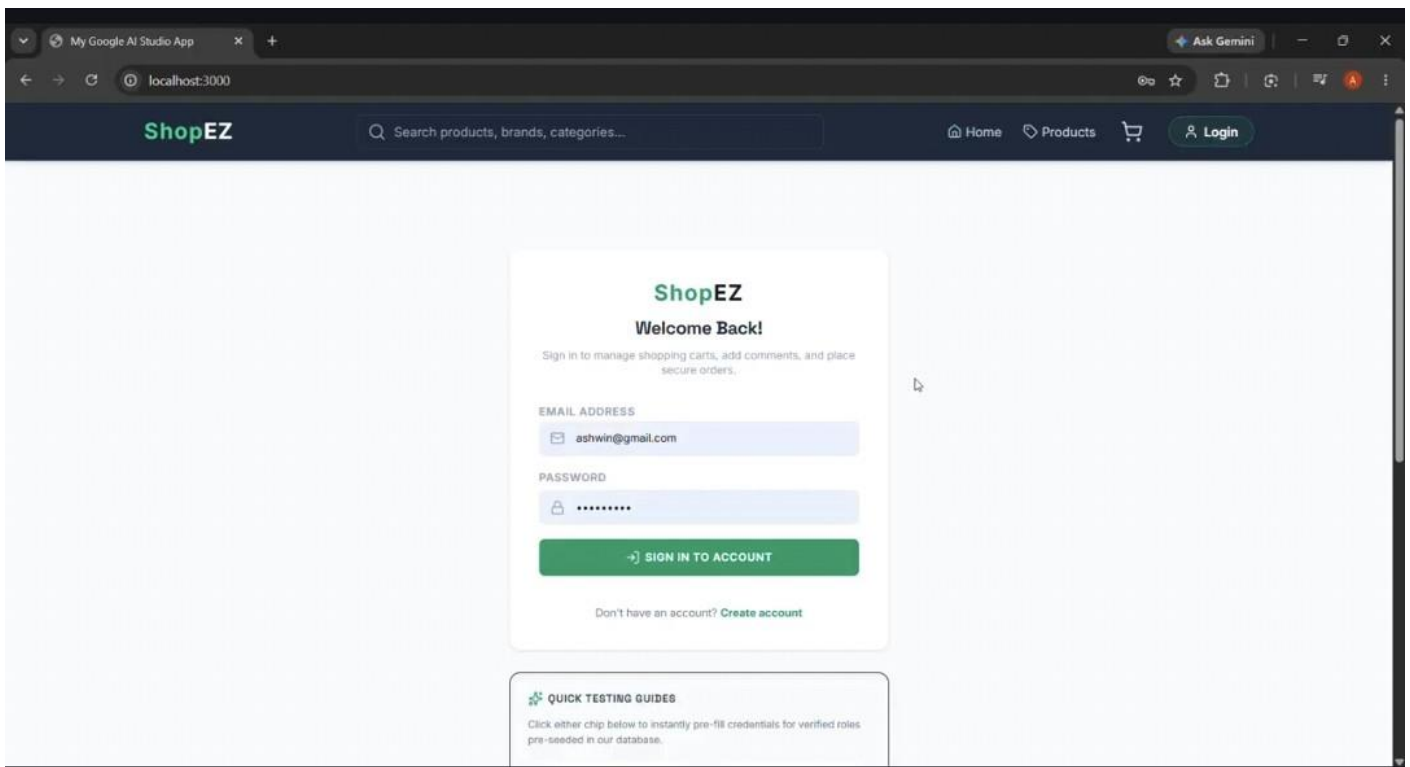
10. Quality Assurance s Testing

A multi-tiered verification strategy was employed to ensure application stability:

- **Unit Testing:** Verified the accuracy of internal logic and data validation utilities.
- **Integration Testing:** Confirmed that Express API endpoints interact correctly with the MongoDB instance.
- **User Scenarios:** Simulated real-world usage, such as verifying the system prevents out-of-stock purchases.

11. Showcase s Repositories

Source Code s Project Link: <https://github.com/m-alladi2006/VIP-C2-SHOPEZ-E-commerce-Application.git>



My Google AI Studio App

Ask Gemini

localhost:3000

ShopEZ

Search products, brands, categories...

HomeProducts

1

Hello, Ramesh

Logout

Return To Shopping Cart

1. Shipping & Logistics Coordinates

RECIPIENT NAME

Ramesh

STREET ADDRESS

1-2-38/1,GOURINATH PURAM

CITY

VAMPUGUDA

POSTAL / ZIP CODE

500062

COUNTRY / REGION

India

CONTACT PHONE NUMBER

09704625093

2. PAYMENT METHODS

☒ Instant Card Authorization

Secure automated card checkout. Decrypt variables seamlessly.

PLACE ORDER AND PAY

CHECKOUT INVOICE

 AeroSound Indigo ANC Headphones

Qty: 1

\$149.99

Subtotal Sum

\$149.99

Shipping fee

FREE

Sales Tax (8%)

\$12.00

Invoice Total

\$161.99

My Google AI Studio App

Ask Gemini

localhost:3000

ShopEZ

Search products, brands, categories...

HomeProducts

Hello, Ramesh

Logout

✓

Order Confirmed!

Thank you for shopping with ShopEZ. Your transaction has processed successfully.

RECEIPT CODE

ord-715-11

SHIPPED VIA

 Expedited Cargo

RECEIPT DETAILS

Ramesh
09704625093

SHIPPING DESTINATION

1-2-38/1,GOURINATH PURAM
VAMPUGUDA, 500062
India

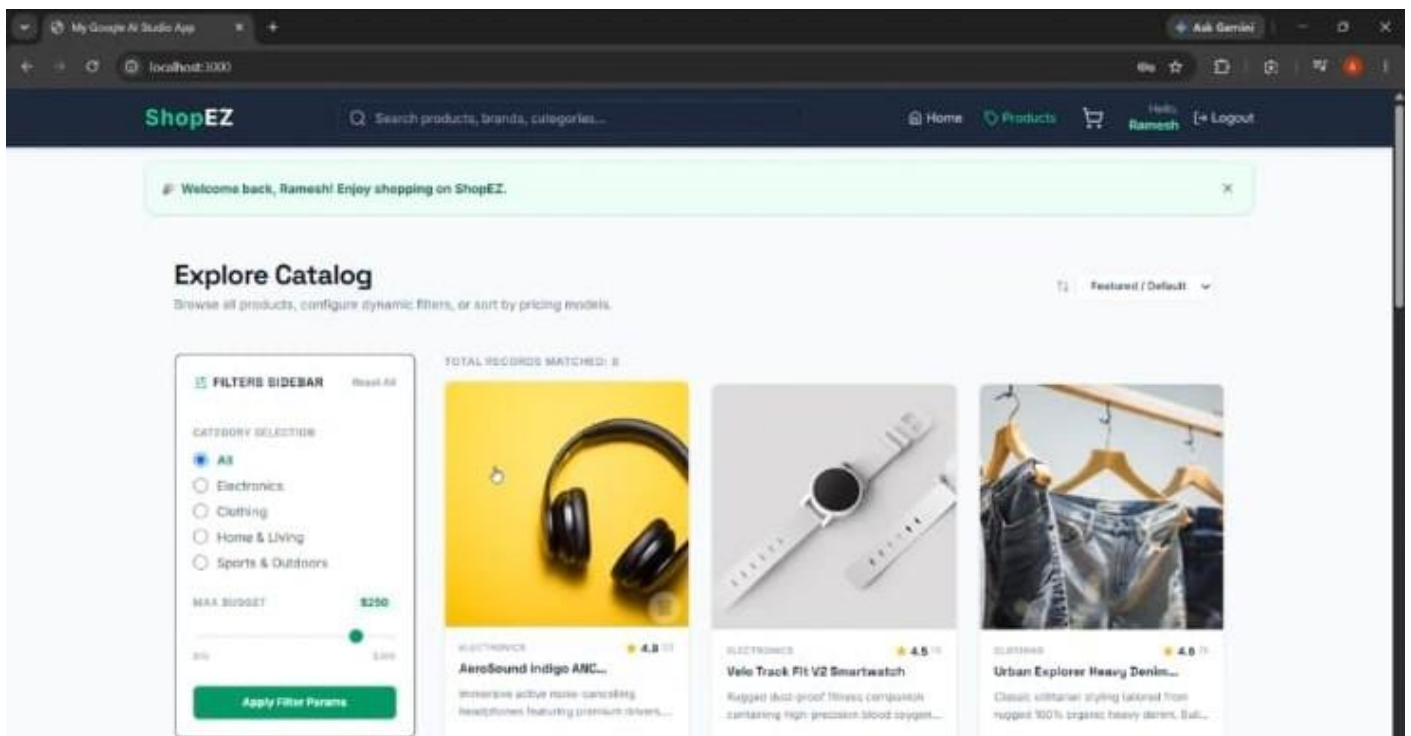
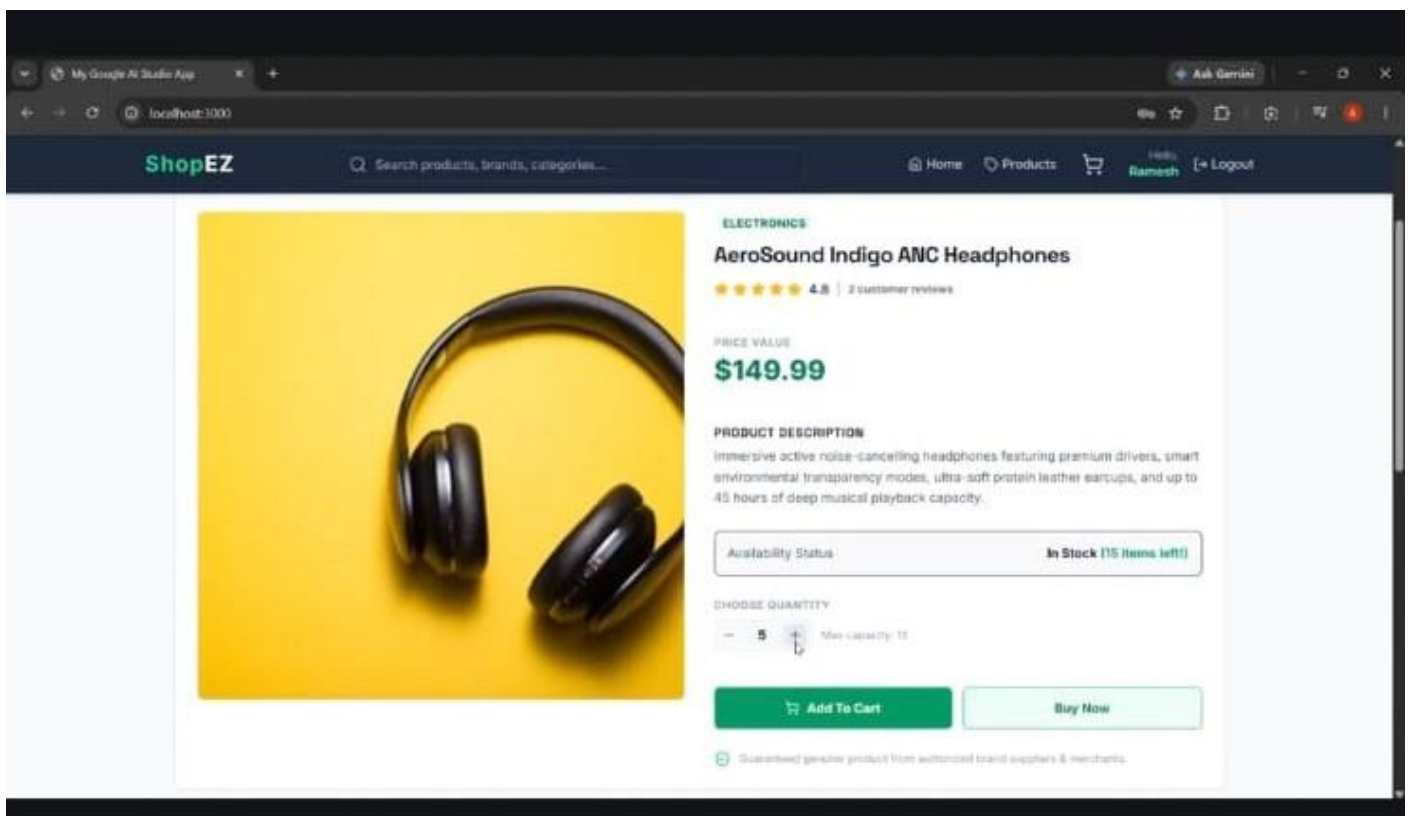
Total Charged Sum

\$161.99

Return to Homepage

Continue retail browsing

Page 9



12. Known Limitations

- Client-side filter states may occasionally require a page refresh for complete synchronization.
- Directly serving massive image files from MongoDB may impact speed; a migration to Cloudinary is currently planned.

13. Roadmap s Future Scope

- Deployment and integration of online payment gateways like Stripe or Razorpay.
- Implementing an AI-driven engine for tailored storefront product recommendations.
- Transitioning the platform into a Progressive Web App (PWA) for native-like offline functionality.